

Exercise 1 :

Consider the following variation of the example from the lecture:

	1	2	3
1			
2			
3			

The only difference is that the reward is 1 point for the gold, -20 points for the snake, and 0 everywhere else.

- Compute the first 3 iterations of value iteration with $\gamma = 0.9$
 - Consider the value function after the 3th iteration. Does the optimal policy change with respect to the version in the lecture? Justify your answer.
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Exercise 2 :

Consider the following variation of the example from the lecture:

	1	2	3
1			
2			
3			

But now the snake will move to an adjacent square every time. The following rules apply:

- If the adventurer is where the snake is, the snake will remain there. In that case, the adventurer cannot move with probability 0.5, else it moves where they wanted to.
- If the adventurer is next to the snake, then the snake will jump to the adventurer with probability 0.5 (and the adventurer cannot move) or remain where it is.
- If the adventurer is not next to the snake, then the snake will move at random (0.25 percent north, 0.25 percent south, and 0.5 remains where it is).
- As in the original problem, when the adventurer moves there is a small chance (20%) that she falls to the right.

Getting the gold provides the adventurer 10 points. Being in the same square as the snake provides -5 points. Everything else provides -0.1 points. Formalize this problem as an MDP. Provide a formal description of the following.

- (i) Set of states
- (ii) Actions
- (iii) Reward function
- (iv) Transition probability function. In particular, describe the possible outcomes for the initial state where we apply the north action.
- (v) Transition probability function. In particular, describe the possible outcomes for the following state where we apply the east action.

	1	2	3
1			
2			
3			

Exercise 3 :

Consider the game of Tetris: [https://en.wikipedia.org/wiki/Tetris_\(NES_video_game\)](https://en.wikipedia.org/wiki/Tetris_(NES_video_game)).

Formalize this problem as an MDP. Provide a description of the following.

- (i) Set of states
- (ii) Actions
- (iii) Reward function
- (iv) Transition probability function
- (v) Initial state
- (vi) Set of terminal states

The description can be just a short text in English textual and not necessarily entirely formal.
